

DESIGN BASIS AND EXECUTIVE SUMMARY

One model. A traceable engineering package.

A compact, independently checked arithmetic example showing how balances, drawings, equipment and economics connect.

10 MW

TOTAL SITE AC INPUT

184 kg/h

HYDROGEN PRODUCTION

1,472 t/year

HYDROGEN AT 8,000 h/year

Defined calculation boundary

- 10.0 MW purchased AC: 9.5 MW to the rectifier and 0.5 MW other auxiliaries.
- 9.2 MW DC at an assumed 50 kWh/kg H₂. Pure H₂ and O₂; no purge or product loss.
- Ideal water-only RO at 75% recovery; all reaction streams referenced to 25°C.
- 8,000 operating hours/year. Steady output and constant real annual costs.

What this example demonstrates

A reproducible chain from stated assumptions to a heat and material balance, tagged process diagram, representative water-interface P&ID, equipment register and transparent unit-cost calculation.

Calculation assurance

Mass and elemental identities, AC/DC and HHV accounting, nine sensitivity cases and an independent discounted-cash-flow cross-check all pass. These checks validate the arithmetic, not equipment performance.

Maturity: illustrative screening reference. No vendor guarantees, site conditions, pressure/purity specification, equipment sizing approval or AACE cost-estimate class is claimed.

REF-HMB-001 | HEAT AND MATERIAL BALANCE

Every headline figure has a calculation basis.

Reference case: 9.2 MW DC, 50 kWh/kg H2, 8,000 h/year. Stream tags are reused on the PFD.

Material balance | kg/h at the stated basis

TAG / STREAM	FLOW kg/h	BASIS
S01 Raw water	2,192.466	S02 / 0.75
S02 RO permeate / consumed H2O	1,644.350	$184 \times 18.01528 / 2.01588$
S03 RO reject	548.117	S01 - S02
S04 H2 product	184.000	$9.2 \times 1,000 / 50$
S05 O2 product	1,460.350	$184 \times 31.9988 / (2 \times 2.01588)$

Electrical and HHV accounting | MW

ACCOUNT	MW	INTERPRETATION
Purchased AC	10.0000	Total input
Rectifier AC / DC	9.5 / 9.2	0.3000 conversion loss
Other auxiliaries	0.5000	Assumed site allowance
H2 chemical output	7.2496	$184 \times 39.4 / 1,000$
Cooling + losses	2.7504	Defined remainder

Mass and elemental checks

S01 = S03 + S04 + S05. S02 = S04 + S05.
Reaction: $\text{H}_2\text{O} \rightarrow \text{H}_2 + \frac{1}{2} \text{O}_2$. Mass and elemental residuals are zero within floating-point precision. Values in tables are rounded.

AC consumption and thermal equivalent

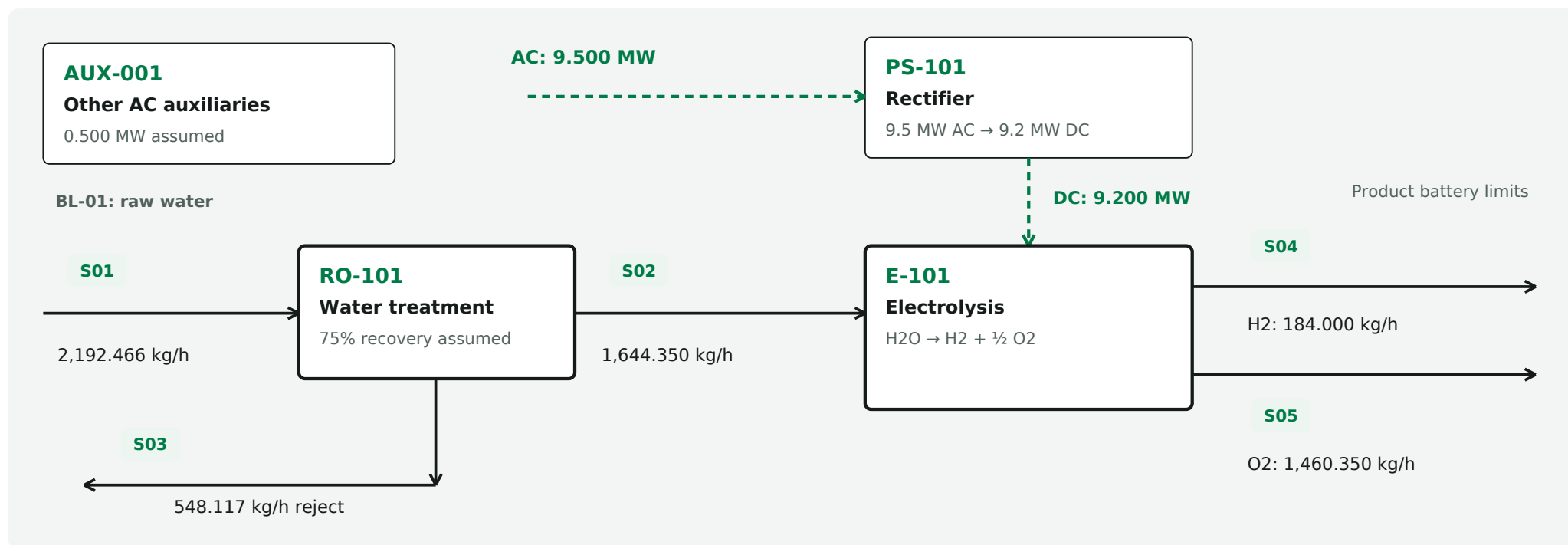
$10,000 / 184 = 54.3478 \text{ kWh/kg H}_2$.
If all 2.7504 MW goes to water at $\Delta T = 10 \text{ K}$: **236.877 m³/h circulation**; not makeup demand.

The HHV remainder is an accounting identity, not an independently predicted cooling duty. Water density 1,000 kg/m³; cp = 4.18 kJ/(kg·K). Additional water treatment losses and cooling makeup are excluded.

REF-PFD-001 | PROCESS FLOW DIAGRAM

The flow diagram and balance use the same tags.

Simplified system boundaries. Product conditioning, pressure, purity and detailed equipment configuration remain undefined.



Drawing conventions

—————> Material stream - - - - -> Electrical connection

RO-101 is an ideal recovery block; E-101 is an aggregate conversion block.
The P&ID illustrates only a candidate water-feed interface within S02.

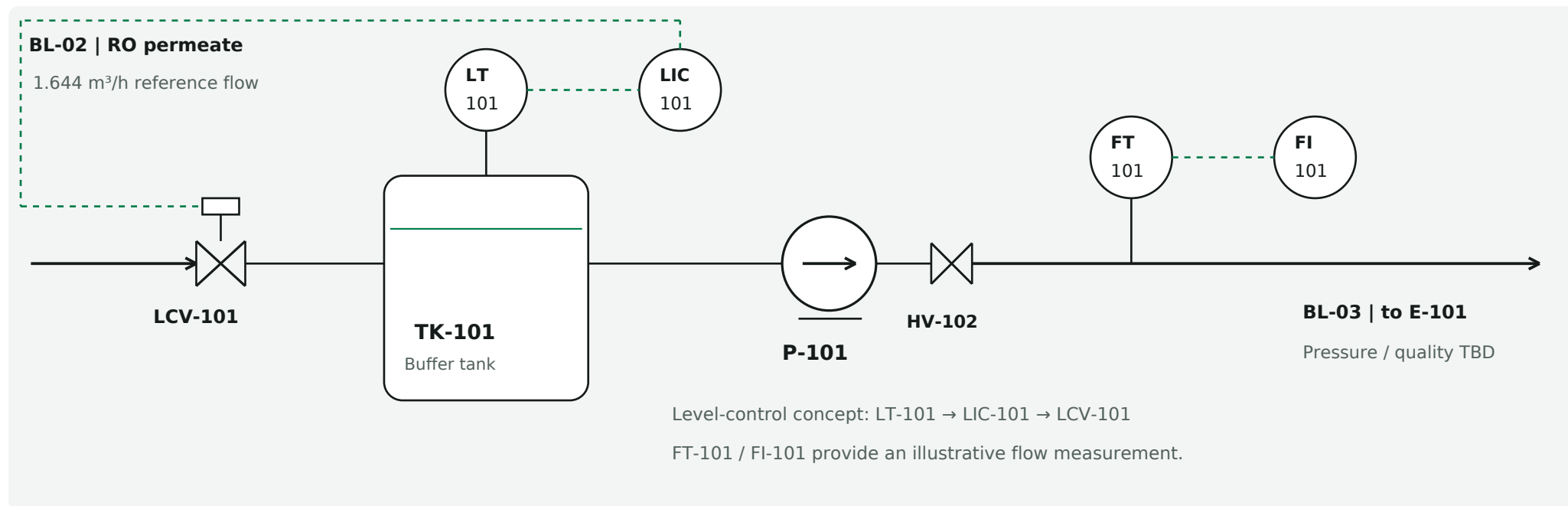
Energy boundary note

10.000 MW site AC = 7.2496 MW H2 HHV + 2.7504 MW combined cooling/losses. The remainder is not shown as a designed utility stream or equipment duty.

REF-PID-001 | REPRESENTATIVE WATER-FEED INTERFACE

Readable engineering intent, with open inputs visible.

Candidate arrangement within S02 only. A preliminary drawing example; no safety function, valve rating or finalized control design is implied.



Legend and scope

Solid: process connection. Green dashed: illustrative signal. LT/LIC: level transmitter/controller. FT/FI: flow transmitter/indicator. HV: manual valve. LCV: level control valve.

Tags are schematic identifiers, not a claim of full ISA/ISO drawing compliance.

Resolve before equipment selection or FEED

Water quality and conductivity limit; delivery pressure; pump head and NPSH; tank working volume; materials; valve fail position; overflow/drain provisions; low-level pump protection; isolation, sampling and required safeguarding.

No line sizes, pressure ratings or safety integrity levels are assigned.

REF-EQR-001 / REF-DS-101 | EQUIPMENT AND DUTY REGISTER

Separate calculated quantities from selection inputs.

A preliminary equipment register records the supported duty basis and the engineering information still required.

TAG / SERVICE	REFERENCE QUANTITY	STATUS / OPEN INPUT
E-101 Electrolysis block	9.200 MW DC; 184 kg H2/h	SEC assumption; technology, pressure and purity TBD
PS-101 Rectifier	9.500 MW AC in; 9.200 MW DC out	96.842% assumed conversion; electrical design TBD
RO-101 Water treatment	2.192 m³/h feed; 1.644 m³/h permeate	75% ideal recovery; raw-water quality / polishing TBD
TK-101 Feed buffer	No volume selected	Hold-up, usable volume, level limits and drains TBD
P-101 Feed pump	1.644 m³/h reference flow	Head, NPSH, motor power and materials TBD
U-CW Thermal equivalent	2.7504 MW; 236.877 m³/h at ΔT 10 K	Conditional remainder; not a validated cooling duty

Sample datasheet | P-101 water-feed pump

FIELD	CURRENT BASIS	FIELD	CURRENT BASIS
Fluid	Process water; quality TBD	Flow	1.644 m³/h reference; margins TBD
Temperature	25°C accounting reference; rating TBD	Head / NPSH	Not established
Materials	To suit defined water chemistry	Drive / control	Motor duty and control details TBD

REF-ECO-001 | ECONOMIC ARITHMETIC AND SENSITIVITY

Reproducible costs. Visible assumptions.

Illustrative EUR values only. No vendor quotation, investment recommendation or cost-estimate classification is implied.

CAPEX €20m | 25 years | 8% real | 8,000 h/year | Power €75/MWh | Raw water €2/m³

Nonpower fixed OPEX: €550,000/year, including assumed consumables and stack reserve; raw water added separately.

Base annualized cost

COMPONENT	€/year	€/kg H2
Annualized capital	1,873,575.58	1.2728
Purchased electricity	6,000,000.00	4.0761
Raw water	35,079.46	0.0238
Nonpower fixed OPEX	550,000.00	0.3736
TOTAL	8,458,655.04	5.7464

$CRF = r(1+r)^n / [(1+r)^n - 1] = 0.093678779/\text{year}.$
 $LCOH = \text{annualized cost} / 1,472,000 \text{ kg/year}.$
Independent discounted cash flows give the same result.

LCOH sensitivity | €/kg H2

Rows: DC SEC (kWh/kg) | Columns: electricity (€/MWh)

DC SEC	50	75	100
45	3.9513	5.1741	6.3969
50	4.3877	5.7464	7.1051
55	4.8241	6.3186	7.8132

Site AC stays 10 MW and DC stays 9.2 MW in every scenario. CAPEX and fixed OPEX stay constant. Changing SEC changes output; water follows stoichiometry. This is sensitivity arithmetic, not a vendor comparison.

Constant real, pre-tax cash flows; no oxygen or heat credit. No separately costed compression, storage, delivery, working capital or site-specific works. Stack-reserve adequacy and degradation are not evaluated.

REF-REG-001 | DELIVERABLE AND HANDOVER REGISTER

Deliverables and open decisions, ready to inspect.

The example can be inspected and rerun. Converting it to a project basis requires verified site, vendor and operating evidence.

ID	DELIVERABLE	NEXT EVIDENCE REQUIRED
REF-HMB-001	Heat and material balance	Confirmed performance maps, losses, purge and temperature/pressure basis
REF-PFD-001	Tagged process flow diagram	Agreed technology, battery limits, gas handling and product specification
REF-PID-001	Representative water-interface P&ID	Water quality, line design, control philosophy and safeguarding review
REF-EQR-001 / DS-101	Equipment register and sample pump sheet	Hydraulic design, sizing margins, materials and vendor duties
REF-ECO-001	Economic calculation and sensitivity	Scope-linked quotations, operating profiles and full cost boundary
reference_model.py / .json	Reproducible calculation source	Validated input register and controlled project revision process

What changes at FEED handover

Freeze battery limits and design cases. Resolve open data. Obtain vendor-backed duties. Define operating and maintenance philosophy. Complete hazards, controls and relevant code reviews. Connect the equipment register to scope-linked costs and procurement requirements.

Source of numbers: task-supplied assumptions, calculated in reference_model.py and independently cross-checked. This reference is a demonstration of deliverable structure and traceability, not a validated client project.

Reference limitations remain explicit

Ideal steady-state chemistry; no gas loss, purge or degradation. Water treatment and auxiliary demand are assumptions. Thermal residual is not equipment sizing. Pressure, purity and product delivery are undefined. Costs are hypothetical and incomplete for investment use.